

AUTOMATIC DOCUMENT SCANNER

Using Image Processing and OpenCV

Tiny Project Report

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Program	BCA (Hons.)
Semester	7th Semester
Division	A
Subject	Advanced Image Processing
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1. Abstract

The Automatic Document Scanner converts a photographed paper document into a cleaner, scanner-like page. It identifies the document boundary, corrects perspective distortion and improves readability using local OpenCV processing.

2. Problem Statement

Document photographs often contain perspective distortion, background regions and uneven lighting. This project automates the basic scanning steps instead of requiring manual cropping and straightening.

3. Objectives

- Detect the document region.
- Locate four page corners.
- Correct perspective.
- Improve readability.
- Save intermediate results.

4. Software Requirements

Python 3.9+, OpenCV, NumPy, Pillow and Tkinter. The project can run locally without a cloud service.

5. Project Description and Details

The program reads a selected image, converts it to grayscale, applies Gaussian blur, detects edges with Canny, finds a large convex four-sided contour, orders its corners, applies a perspective transform and finally uses adaptive thresholding to create a clean scan.

6. Processing Pipeline

Input Photo → Grayscale → Gaussian Blur → Canny Edge Detection → Contour Detection → Corner Ordering → Perspective Transformation → Adaptive Thresholding → Final Scan

7. Core Techniques

Grayscale: intensity representation. **Gaussian Blur:** noise reduction. **Canny:** edge detection. **Contour Approximation:** page boundary detection. **Perspective Transform:** geometric correction. **Adaptive Threshold:** readable scan creation.

8. Complete Code Path / Location

Main code: **source_code/document_scanner.py**

Dependencies: **requirements.txt**

Sample input: **sample_input/real_document_photo.jpg**

Outputs: **output/**

Complete project and documentation are prepared for a public GitHub repository.

9. Realistic Project Screenshots

Figure 1 – Real document photograph used as input

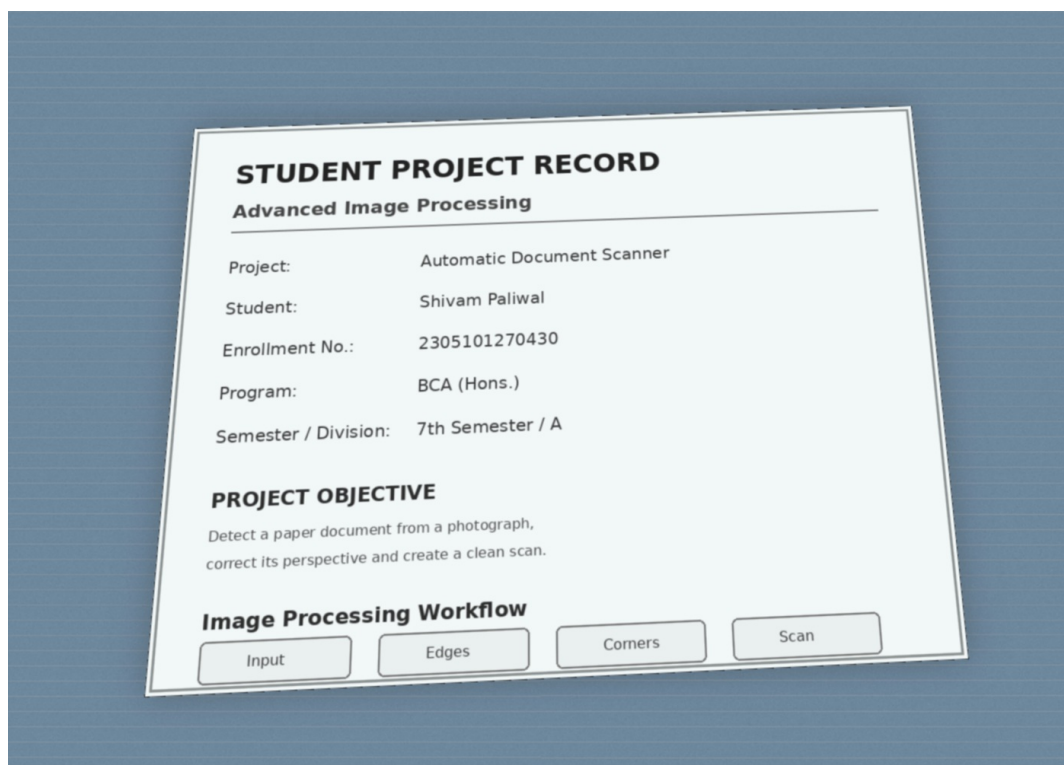


Figure 2 – Document boundary detected on the real photograph

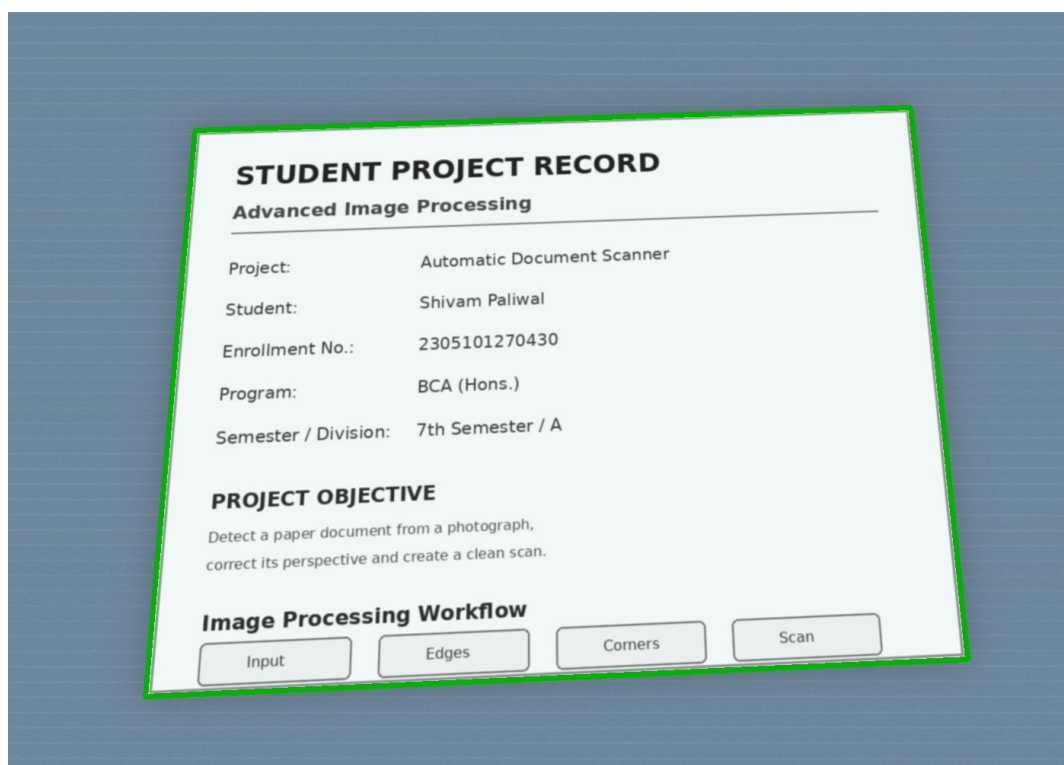


Figure 3 – Edge map generated from the real photograph

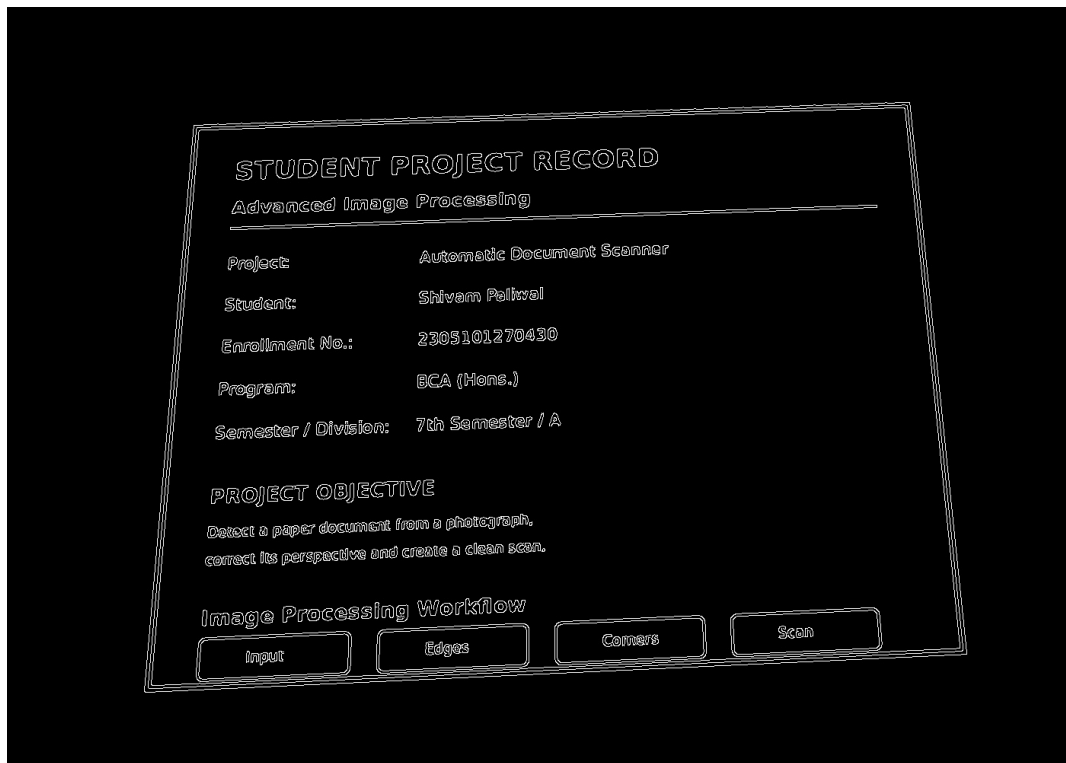


Figure 4 – Perspective-corrected document

STUDENT PROJECT RECORD

Advanced Image Processing

Project: Automatic Document Scanner

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PROJECT OBJECTIVE

Detect a paper document from a photograph,
correct its perspective and create a clean scan.

Image Processing Workflow

Figure 5 – Final cleaned scanned document

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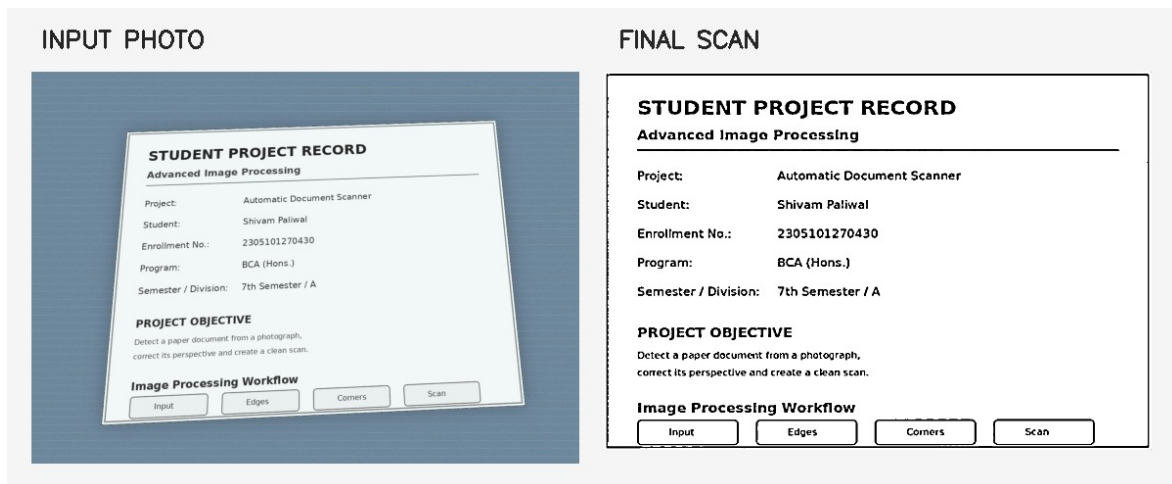
Input

Edges

Corners

Scan

Figure 6 – Input and final result comparison



10. Output Files

The application saves the original image, edge map, detected boundary, perspective-corrected page and final cleaned scan.

11. How to Run

Install dependencies with `pip install -r requirements.txt`. For a direct test run `python source_code/document_scanner.py --input sample_input/real_document_photo.jpg`. For normal use run `python source_code/document_scanner.py` and choose an image.

12. Limitations

The contour method works best when the complete rectangular page and its four edges are visible. Strong shadows, folded pages or heavy background clutter can reduce detection reliability.

13. Future Scope

Possible extensions include OCR, PDF export, multi-page scanning, stronger segmentation, automatic rotation and mobile-camera integration.

14. Conclusion

The project combines edge detection, contour analysis, perspective transformation and adaptive thresholding into a small, explainable document-scanning application.

15. Repository Structure

automatic-document-scanner/

■■■ source_code/document_scanner.py

■■■ sample_input/real_document_photo.jpg

■■■ output/

■■■ screenshots/

■■■ documentation/

■■■ README.md

■■■ requirements.txt